Reference angles and trigonometric functions

1
- **a** Quadrant 2
- **b** 45°
- **c** Positive
- **d** $\sin 45^\circ$

2
- **a** Quadrant 3
- **b** 60°
- **c** Negative
- **d** $-\cos 60^\circ$

3
- **a** Quadrant 3
- **b** 30°
- **c** Positive
- **d** $\tan 30^\circ$

4
- **a** Quadrant 3
- **b** $\frac{\pi}{6}$
- **c** Negative
- **d** $-\sin \frac{\pi}{6}$

5
- **a** Quadrant 4
- **b** $\frac{\pi}{6}$
- **c** Positive
- **d** $\cos \frac{\pi}{6}$

6
- **a** Quadrant 3
- **b** $\frac{\pi}{4}$
- **c** Positive
- **d** $\tan \frac{\pi}{4}$

7
- **a** Quadrant 4
- **b** $\frac{\pi}{6}$
- **c** Negative
- **d** $-\tan \frac{\pi}{6}$

8
- **a** Quadrant 4
- **b** $\frac{\pi}{4}$
- **c** Positive
- **d** $\cos \frac{\pi}{4}$

9
- **a** Quadrant 3
- **b** $\frac{\pi}{3}$
- **c** Negative
- **d** $-\sin \frac{\pi}{3}$

10
- **a** Quadrant 3
- **b** $\frac{\pi}{3}$
- **c** Negative
- **d** $-\cos \frac{\pi}{3}$

11
- **a** Quadrant 4
- **b** $\frac{\pi}{4}$
- **c** Negative
- **d** $-\tan \frac{\pi}{4}$

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Exact values of trigonometric functions

12

a $(0, 1)$

b 0

c 1

d

i No **ii** No **iii** Yes **iv** Yes

13

a $(1, 0)$

b 1

c 0

d

i Yes **ii** Yes **iii** No **iv** No

14

a $(0, -1)$

b 0

c -1

d

i Yes **ii** No **iii** No **iv** Yes

15

a $(-1, 0)$

b -1

c 0

d

i No **ii** Yes **iii** Yes **iv** No

16

a $(1, 0)$

b 1

c 0

d

i No **ii** Yes **iii** No **iv** Yes

17

a $(-1, 0)$

b -1

c 0

d

i Yes **ii** Yes **iii** No **iv** No

18

a $-\frac{1}{2}$ **b** $\frac{\sqrt{3}}{2}$ **c** $-\frac{\sqrt{3}}{3}$ **d** -2

e $\frac{2\sqrt{3}}{3}$ **f** $-\sqrt{3}$ **g** $\frac{1}{2}$ **h** $-\frac{\sqrt{3}}{2}$

i $-\frac{1}{\sqrt{3}}$ **j** 2 **k** $-\frac{2\sqrt{3}}{3}$ **l** $-\sqrt{3}$

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a $\frac{\sqrt{3}}{2}$ b $-\frac{1}{2}$ c $-\sqrt{3}$ d $\frac{2\sqrt{3}}{3}$

e -2 f $-\frac{\sqrt{3}}{3}$

20

a i 60° ii $-\frac{1}{2}$ b i 0 ii 0 c i $\frac{\pi}{3}$ ii $-\frac{1}{2}$

21

a 1 b 1 c -1 d -1

e Undefined f Undefined g $\sqrt{3}$ h $\frac{2\sqrt{3}}{3}$

i $\frac{2\sqrt{3}}{3}$ j $\sqrt{2}$ k $\sqrt{2}$ l 1

m 0 n -1 o Undefined p 0

q Undefined r Undefined

22

a 1 b $\frac{-1}{2}$ c $\frac{1}{2}$ d 1

e $\frac{1}{2}$ f $-\frac{1}{\sqrt{2}}$ g $-\frac{1}{2}$ h $-\frac{\sqrt{3}}{2}$

i $\frac{1}{\sqrt{3}}$ j $-\frac{\sqrt{3}}{2}$ k $\frac{1}{2}$ l -1

m $-\frac{1}{\sqrt{3}}$ n Undefined o Undefined p Undefined

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a $-\frac{1}{4}$ b $-\frac{\sqrt{3}}{4}$ c -2 d $-2\sqrt{3}$

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